

Pore-Scale Study of Transverse Mixing Induced CaCO_3 Precipitation and Permeability Reduction in a Model Subsurface Sedimentary System

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A microfluidic pore structure etched into a silicon wafer was used as a two-dimensional model subsurface sedimentary system (i.e., micromodel) to study mineral precipitation and permeability reduction relevant to groundwater remediation and geological carbon sequestration. Solutions containing CaCl_2 and Na_2CO_3 at four different saturation states ($\Omega = [\text{Ca}^{2+}][\text{CO}_3^{2-}]/K_{\text{spCaCO}_3}$) were introduced through two separate inlets, and they mixed by diffusion transverse to the main flow direction along the center of the micromodel resulting in CaCO_3 precipitation. Precipitation rates increased and the total amount of precipitates decreased with increasing saturation state, and only vaterite and calcite crystals were formed (no aragonite). The relative amount of vaterite increased from 80% at the lowest saturation state ($\Omega_v = 2.8$ for vaterite) to 95% at the highest saturation state ($\Omega_v = 4.5$). Fluorescent tracer tests conducted before and after CaCO_3 precipitation indicate that pore spaces were occluded by CaCO_3 precipitates along the transverse mixing zone, thus substantially reducing porosity and permeability, and potentially limiting transformation from vaterite to the more stable calcite. The

results suggest that mineral precipitation along plume margins can decrease both reactant mixing during groundwater remediation, and injection and storage efficiency during CO_2 sequestration.

Introduction

Geological carbon sequestration is being evaluated at several field sites around the world to determine its technical feasibility, economical viability for storing carbon produced from stationary sources (e.g., refs 1–5) and the ability to mitigate climate change (6). Carbon dioxide capture, transport, and storage (CCS) (7) involves the capture of CO_2 from fossil fuel combustion, compression of captured CO_2 into a liquid or supercritical fluid state, transport and subsequent injection. The most volumetrically significant subsurface rock targets for injection are deep subsurface saline aquifers, depleted oil reservoirs, and unmineable coal seams (8). Injected CO_2 displaces brine or oil in pores and also rises vertically toward the caprock, being less buoyant than the saline brines housed within subsurface sedimentary reservoirs. Along fluid–fluid interfaces between injected CO_2 and pore water, CO_2 dissolves into the aqueous phase and either reduces the pH, or reacts with species that buffer pH to increase bicarbonate and carbonate concentrations (9, 10). Decreasing pH conditions induce carbonate mineral dissolution, which release divalent metal cations, and increase bicarbonate and carbonate concentrations. Simulation results indicate that these conditions promote carbonate mineral precipitation along the margins of injected CO_2 plumes (11–13). Two significant concerns associated with mineral dissolution and precipitation during CO_2 injection are dissolution and failure of the caprock seal, and precipitation of minerals that reduce permeability, reservoir capacity, and CO_2 injection efficiency.

A large body of literature has explored dissolution and precipitation reactions relevant to CO_2 injection in laboratory batch studies (e.g., refs 14–16). Which mineral species react and the time scales of reaction depend on a number of factors including existing minerals and solutes, pH, temperature, and pressure (17–19). Among the most common minerals available for reaction in geologic storage reservoirs are calcium carbonates, which include vaterite, aragonite, and calcite (20). They are often present in limestone in sedimentary deposits, and dissolution and precipitation time scales are rapid (i.e., minutes to days). Carter et al. (20) showed that in a batch system, vaterite is initially the major phase at low temperature (i.e., 20 °C), and aragonite is initially the major phase at temperatures above 50 °C; both transformed to the more stable form calcite after 7 h.

Although calcium carbonate dissolution and precipitation reactions have been extensively studied in batch systems (e.g., refs 9, 15, 16, and 20), very little work has explored these reactions under flow conditions, where advective and diffusive mixing processes play a major role (21, 22). Preliminary modeling studies (11–13) indicate that along the fringes of injected CO_2 plumes, convective and diffusive mixing processes affect CO_2 dissolution into the pore water and subsequent dissolution and precipitation reactions within the formation and caprock. Calcium carbonate precipitation can decrease porosity and permeability. This has been shown in wellborn cement and caprock core samples (23), and in controlled two-dimensional (2D) benchtop flow cell experiments where calcium carbonate was precipitated along the center mixing line between aqueous phase calcium and carbonate (21, 22). Calcium

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TABLE 1. Summary of Experimental Conditions

exp	conc. [mM]	CaCl ₂ pH ^a	Na ₂ CO ₃ pH ^a	Ω _c wrt calcite ^b	Ω _a wrt aragonite ^c	Ω _v wrt vaterite ^d
1	6.5	5.8	10.6	3.4	3.2	2.8
2	10	5.9	10.8	3.8	3.6	3.1
3	25	6.1	10.9	4.6	4.4	3.9
4	50	6.3	11.1	5.2	5.0	4.5

^a pH of DI water used to prepare all solutions is 5.7. ^b $pK_{sp} = 8.42$. ^c $pK_{sp} = 8.22$. ^d $pK_{sp} = 7.73$; all pK_{sp} values from Stumm (36).

carbonate precipitation limited by advective or diffusive mixing is relevant to other subsurface processes as well, including precipitation near injection wells and along plume margins during groundwater remediation (24), precipitation along fluid injection fronts during enhanced oil recovery (25), and precipitation of leachate along the shadow zones of nuclear waste storage tunnels (26).

Herein, we focus on calcium carbonate precipitation and dissolution under transverse mixing limited conditions. Reactive transport limited by transverse mixing has been the focus of a number of recent studies. However, in all but a few cases reactions of interests were limited to soluble reaction products (e.g., refs 27–30). Transverse mixing-limited reactive transport involving mineral precipitation and dissolution has been limited to numerical simulations (e.g., refs 21, 22, and 31–34) and two studies in continuum-scale flow cells (e.g., refs 21 and 22). These studies all indicate that mixing limitations between dissolved cations and carbonate give rise to highly nonuniform concentration gradients, localized reactions and precipitation, and hence highly localized changes in porosity and permeability. However, experimental pore-scale studies are lacking, and the influence of these conditions on mineral growth rate, morphology, phases, and pore blockage are unknown.

The objective of this study is to investigate calcium carbonate precipitation limited by transverse mixing between calcium chloride and sodium carbonate in the context of flow through porous subsurface sedimentary systems. Specifically, we evaluate the impact of transverse concentration gradients and solution saturation state on calcium carbonate precipitate morphology, phase, stability, and reaction rate in a homogeneous pore network etched into a silicon substrate (i.e., micromodel). Reflected differential interference contrast microscopy is used to evaluate the precipitate morphology, stability, and reaction rate, and confocal Raman spectroscopy is used to evaluate the precipitate phase. Micromodel flow and transport systems are excellent platforms for elucidating coupled processes involving advection, diffusion, and biogeochemical reactions at the pore-scale, and for providing insights to continuum-scale processes (30, 35). While reactant species other than calcium and carbonate are also important in subsurface sedimentary systems, this research focuses on these as the first step in evaluation of coupled pore-scale physical processes (advection, diffusion) and geochemical reactions that potentially impact CO₂ sequestration and other subsurface processes.

Materials and Methods

Chemicals. Solutions of calcium chloride (CaCl₂), sodium carbonate (Na₂CO₃), and hydrochloric acid (HCl) (all chemicals are ACS reagents grade, Sigma-Aldrich) were prepared in deionized (DI) water and degassed under sonication prior to use. Four equal molar concentrations of CaCl₂ and Na₂CO₃ were prepared: 6.5, 10, 25, and 50 mM. Table 1 lists the pH and saturation state (Ω) of each set of solutions with respect to (wrt) three different polymorphs of CaCO₃ (i.e., calcite, aragonite, and vaterite) (36). The saturation state is defined


FIGURE 1. Micromodel and flow setup.

as $\Omega = [\text{Ca}^{2+}] \times [\text{CO}_3^{2-}] / K_{sp\text{CaCO}_3}$ (37); concentrations of Ca²⁺ and CO₃²⁻ are based on the inlet reactant concentrations. The experimental conditions were chosen as a first step to evaluate mineral precipitation rates, morphology, phase, and the corresponding effects on permeability; they do not directly reflect the range expected in an actual reservoir.

Fluorescein (Sigma, 70%) was dissolved in DI water at 10 μM and used as a fluorescent tracer to visualize changes in permeability as a result of CaCO₃ precipitation.

Micromodel Experiments. The micromodel (Figure 1) used in this study consists of two separate inlet channels (A, B) connected to a 2 cm × 1 cm (length × width) pore network, followed by a third inlet (C) and an outlet (D). The pore network contains a uniform distribution of cylindrical posts 300 μm in diameter, with 180 μm pore spaces, 40 μm pore throats, and a porosity of 0.39. The depth of the pore network is 17–21 μm. Standard photolithography techniques and ICP-DRIE (inductively coupled plasma-deep reactive ion etching) were used to fabricate the micromodels; details of the microfabrication process can be found in Chomsurin and Werth (38) and Willingham et al. (29). The fabricated silicon micromodels were heated to 1100 °C under oxygen for 1 h to grow a thin oxide layer (~0.1 μm). The final step was to anodically bond a cover sheet of Pyrex glass to etched silicon micromodel to seal the flow channels.

Prior to each experiment, the micromodel was first flushed with CO₂ gas to facilitate subsequent saturation by water, followed by >100 pore volumes (PV) of degassed DI water. Solutions of CaCl₂ and Na₂CO₃ were introduced separately into the pore network via inlets A and B at equal flow rates corresponding to a Darcy velocity of 1.25 cm/min; constant flow through the micromodel was maintained using a pair of ISCO pumps (Teledyne ISCO Inc., model 100DM). To prevent CaCO₃ precipitation and clogging in the outlet tubing, a solution of HCl (pH = 1.5–2.0) was introduced through inlet C using a syringe pump (Harvard Instruments) at a volumetric flow rate equal to one-half of the total flow rate through the pore network. An independent test using a fluorescent tracer showed minimal flow disruption in the pore network due to injection through C. All experiments were performed in duplicate, and with previously unused micromodels.

Microscope Imaging. Mixing between the two reactant solutions resulted in calcium carbonate precipitation in the center of the micromodel, which was imaged at selected time intervals until precipitation reached pseudo steady state (i.e., the number of pixels occupied by CaCO₃ precipitates in a micromodel was constant over time). All micromodel images were acquired with a Nikon Eclipse 2000 TE epifluorescent microscope using 4× and 10× inverted objectives (with image resolutions of 1.62 and 0.65 μm, respectively) and reflected differential interference contrast (DIC) microscopy. The microscope is equipped with an automated stage (Prior Scientific Instrument) and a monochrome digital Charge-Coupled Device (CCD) camera, both connected to a computer and controlled by NIS-Elements imaging software. A total of 75 (25 × 3) separate images were taken at each time point; an additional image from outside the pore network area was also acquired and applied to each image to correct for nonuniform illumination. Next, the 25 × 3 separate images

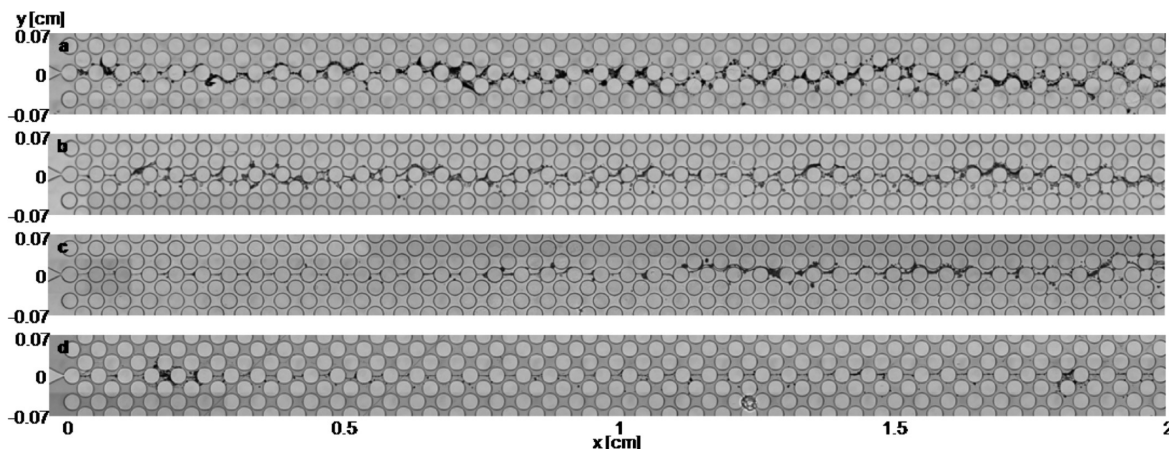


FIGURE 2. Images of center of micromodels with CaCO_3 precipitates formed along the mixing zone at different saturation states (a) $\Omega_c/\Omega_v = 3.4/2.8$, (b) $\Omega_c/\Omega_v = 3.8/3.1$, (c) $\Omega_c/\Omega_v = 4.6/3.9$, and (d) $\Omega_c/\Omega_v = 5.2/4.5$.

were montaged to form a single image, which captured the center of the pore network where CaCO_3 precipitates were formed.

Image Analysis. Montaged images ($0.65 \mu\text{m}$ resolution) of micromodels were analyzed to quantitatively evaluate CaCO_3 precipitation rates. The image intensity of CaCO_3 precipitates is lower than that for silicon posts and water-filled pore spaces; a threshold value was determined for each image to distinguish pixels with CaCO_3 precipitates, and the area (i.e., number of pixels) corresponding to precipitates was determined. This method only results in an apparent indicator of precipitation rate, since precipitates may not occupy the entire depth of the micromodel, and because the density of precipitates may vary between pixels.

Raman Back-Scattering Spectroscopy. This method was used to characterize calcium carbonate mineral phases in selected pores along the mixing zone when precipitation in a micromodel reached pseudo steady state. A long working distance $50\times$ objective was used to locate precipitates in the micromodel, to help focus 30 mW of 514.0 nm polarized excitation from a CW Nd:YAG diode laser to an approximate $1.5 \mu\text{m}$ spot size and to collect the Raman scattered light from the precipitate. Video imaging of the sample under laser illumination allowed precise positioning of the focused laser beam. The backscattered Raman light was collected and focused at the $200 \mu\text{m}$ entrance slit of a JY 800 triple spectrometer with 800 g/mm gratings. The exit slit was maintained at $200 \mu\text{m}$ and the scattered light was then dispersed on to a liquid nitrogen cooled charged coupled device Ge detector. The detector was calibrated using Hg emission lines. Since the Raman spectrum of CaCO_3 polymorphs ranges from about 100 to 1600 cm^{-1} , collection of the entire spectrum in two segments is necessary. The Raman signal was averaged for a minimum of 90 s for each segment. Multiple Raman spectra were collected on the three morphologically distinct crystallite types, and the peak positions and lineshapes of the resulting spectra were analyzed using a commercially available software package, GRAMS/32. Since Raman spectra reflect the molecular symmetry of the crystalline lattice, the pattern of peaks can be used to distinguish between polymorphs of CaCO_3 (20). The collected Raman spectra were assigned to either vaterite or calcite crystal structure by comparison to both a spectral database and spectra reported in literature (20, 39, 40).

Fluorescent Tracer Experiments. Immediately after CaCO_3 precipitation reached pseudo steady state in a micromodel, a fluorescein tracer solution and water were separately injected into the micromodel through inlets A and B, at a 2:1 flow rate ratio. A separate control experiment was performed using the same flow rates in a micromodel with no precipi-

tates. By comparing the flowpath of fluorescein in micromodels with and without precipitates, permeability changes due to precipitation were identified.

Laser Confocal Imaging. Also after precipitation reached pseudo steady state, the micromodel pore network was entirely filled with the fluorescein solution and fluorescent confocal images were obtained using a Leica SP2 laser scanning confocal microscope at $0.21 \mu\text{m}$ in-plane resolution and $0.5 \mu\text{m}$ slice thickness to visualize the three-dimensional (3D) morphology of CaCO_3 precipitates.

Results and Discussions

Distribution, Morphology, and Mineralogy of CaCO_3 Crystal Precipitation. Images of micromodel pore spaces with calcium carbonate precipitates at pseudo-steady-state in each of the four experiments are shown in Figure 2. At all saturation states, crystal precipitation took place primarily in pore spaces along the center mixing zone between calcium and carbonate, with some precipitates forming around the edges of cylindrical posts. The transverse width of precipitates decreases with increasing saturation state. This zone is approximately two pore bodies wide at the two lower saturation states (Figure 2a,b), and less than one pore body at the two higher saturation states (Figure 2c,d). The total amount of CaCO_3 precipitation formed (based on the total number of pixels containing precipitate) decreases with increasing saturation state, with this amount approximately 5 times greater at the lowest saturation state than at the highest saturation state. For water-soluble species, higher saturation states are expected to result in more reactant mixing (greater concentration driving force), and more product formation. The opposite is true for CaCO_3 precipitation in the micromodels. This indicates that precipitates formed at higher saturation states are more densely packed (i.e., CaCO_3 precipitates have lower porosity) and limit transverse mixing. At pseudo-steady-state, the width of each precipitate zone in the transverse direction remained nearly constant along the flow direction in all experiments. This agrees with results from a continuum-scale study (21) in which mixing of calcium chloride and sodium carbonate in a 2D flow cell resulted in a calcium carbonate precipitate zone with uniform width.

Images of CaCO_3 precipitates in selected pore spaces at different saturation states, along with Raman spectra of selected CaCO_3 precipitates, are shown in Figure 3. At the lowest saturation state (i.e., $\Omega_c/\Omega_v = 3.4/2.8$ wrt calcite/vaterite, no aragonite detected), three morphologically distinct types of crystals precipitated along the center mixing zone as shown in Figure 2. These include (1) euhedral $50\text{--}100 \mu\text{m}$ diameter rhombohedra (Figure 3a inset) that exhibit distinct growth hillocks (41), (2) feather “shrub” aggregates

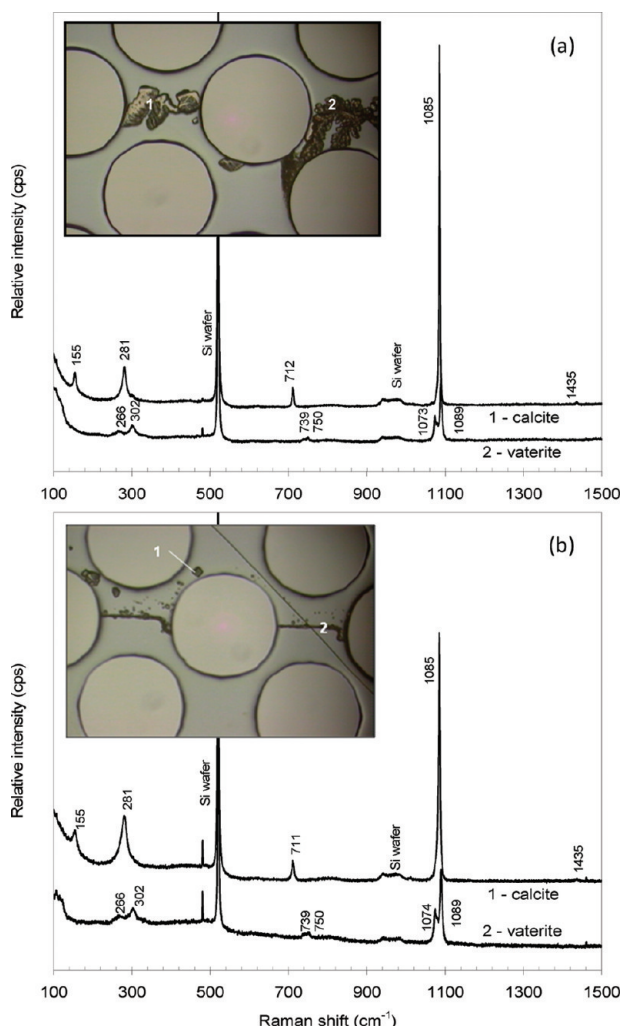


FIGURE 3. Raman back scattering spectra of CaCO_3 precipitates formed along the center mixing zone in the micromodel at (a) $\Omega_c/\Omega_v = 3.4/2.8$ and (b) $\Omega_c/\Omega_v = 5.2/4.5$.

(42, 43) composed of dendritically branched euhedral to subhedral columnar crystals, each of which reach 25 to 100 μm in length along the C-axis (Figure 3a inset), and (3) anhedral to spherical inclusion-rich microcrystalline crystals 5–20 μm in diameter (data not shown). Raman spectra of rhombohedra (Figure 3a) show medium to strong peaks at wavenumbers 155, 281, 712, and 1085 cm^{-1} that match those for calcite (20). Raman spectra of feather “shrub” aggregates (Figure 3a) and spherical microcrystalline crystals show medium to strong peaks at wavenumbers 266 and 302 (lattice/external mode), and 1073 and 1089 (internal mode) that match vaterite (20, 39, 40). Assuming crystals are uniform over the depth of the micromodel, precipitate shapes were used to estimate mineral forms, and vaterite and calcite accounted for approximately 80% and 20%, respectively, of the precipitates (by area).

At the highest saturation state (i.e., $\Omega_c/\Omega_v = 5.2/4.5$), the majority of precipitates were confined to thin dense zones distributed across pore bodies and around the perimeters of cylindrical posts along the center mixing zone, with isolated crystalline precipitates (both rhombohedra and spherical microcrystal) formed off center along the walls of cylindrical posts (Figure 3b inset). Raman spectra (Figure 3b) of solid precipitate lines and spherical microcrystals matched vaterite as characterized by peaks at wavenumbers 266, 302, 1074, and 1089 (20, 39, 40). Rhombohedra were found to be calcite. On the basis of the precipitate shapes, greater than 95% of the total precipitates were vaterite, with the remainder (<5%)

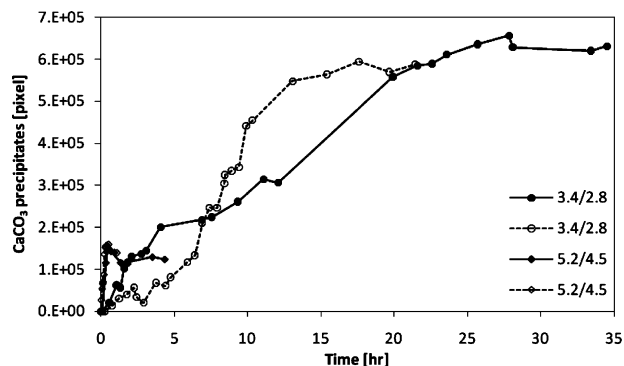


FIGURE 4. Overall CaCO_3 precipitation versus time in the micromodel at $\Omega_c/\Omega_v = 3.4/2.8$ and $\Omega_c/\Omega_v = 5.2/4.5$.

calcite. The amounts of vaterite and calcite for the intermediate saturation states were in between those in the low and high saturation state experiments.

Vaterite has the highest solubility among the three CaCO_3 polymorphs at ambient temperature and pressure, and is considered a metastable form that transforms to calcite (the more stable form) through Ostwald ripening. Carter et al. (20) showed that at 20 $^\circ\text{C}$, calcite accounts for <5% of the total precipitates in batch system after 2 min, and the percentage increased to 10–30% after 7 h. In the micromodel, Ca^{2+} and CO_3^{2-} are initially segregated and they only mix through diffusion. Their concentrations vary in space and Ω is at a maximum along the micromodel center zone, where CaCO_3 precipitates are preferentially formed. At high saturation state, we propose that rapid precipitation reduced mixing between Ca^{2+} and CO_3^{2-} and hence slowed phase transformation from vaterite to calcite.

CaCO_3 Precipitation Rate. The total number of pixels with CaCO_3 precipitate in the micromodel over time in the low and high saturation state experiments is shown in Figure 4 (results from intermediate saturation states are included in Figure S2 of the Supporting Information). From low to high saturation states, CaCO_3 precipitates initially formed at apparent rates of approximately 3.1×10^4 , 1.1×10^5 , 2.8×10^5 , and 4.0×10^5 pixels/h, and reached pseudo-steady-state after 25, 7, 1.5, and 0.5 h. Assuming uniform precipitate density in each pixel, and a bulk density of 2.6 g/mL (44), the precipitates formed at rates of approximately 6.8×10^{-7} , 2.5×10^{-6} , 6.0×10^{-6} , and 8.9×10^{-6} g/h, and the total amount of precipitates formed are 1.4×10^{-5} , 9.7×10^{-6} , 4.0×10^{-6} , and 2.7×10^{-6} g (from low to high saturation states). The increase in saturation state resulted in higher reactant concentrations in the mixing zone, and hence an increase in precipitation rate and a decrease in the amount of precipitates. It is important to note that precipitate density is likely not spatially uniform. However, the monotonic trends in precipitation rates and total precipitates formed at four different saturation states indicate changes in precipitate density would not change our general results.

The number of pixels containing CaCO_3 precipitate in three pores over time located upgradient, middle, and downgradient along the flow direction in the low and high saturation experiments are shown in Figure 5; results from the two intermediate saturation experiments are included in Figure S3 of the Supporting Information. At upgradient pores near the inlet, CaCO_3 precipitates were formed at higher rates (i.e., line slopes are greater) than the middle and downgradient pores; for example, at $\Omega_c/\Omega_v = 3.4/2.8$, the amount of precipitates in the upgradient pore space reached a maximum value after approximately 7.5 h, whereas in pores near the middle and downgradient sections, CaCO_3 precipitates reached a maximum after 20 h (Figure 5a). This behavior is general for the other two saturation states. The higher precipitation rate in the upgradient pores can be attributed

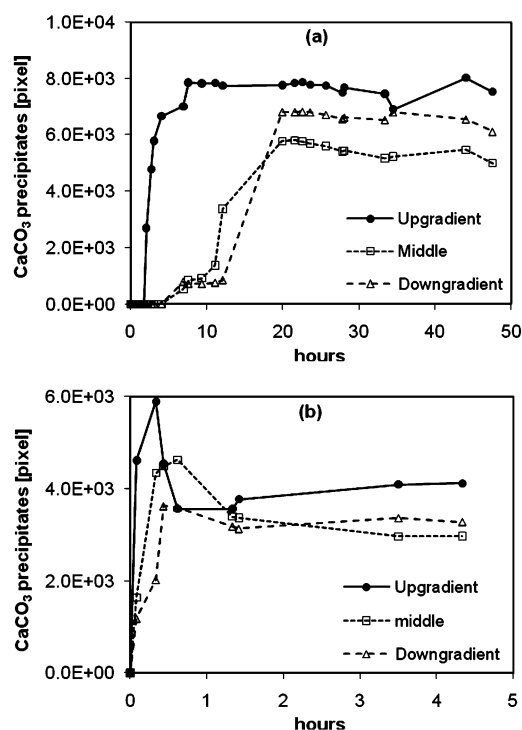


FIGURE 5. CaCO_3 precipitation versus time in selected pores at (a) $\Omega_c/\Omega_v = 3.4/2.8$ and (b) $\Omega_c/\Omega_v = 5.2/4.5$.

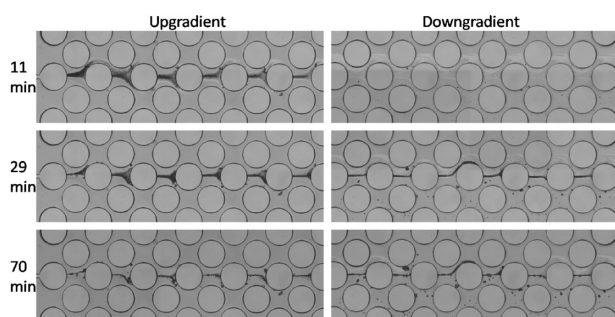


FIGURE 6. CaCO_3 precipitation and dissolution ($\Omega_c/\Omega_v = 5.2/4.5$).

to sharper reactant concentration gradients. Further, at high saturation state ($\Omega_c/\Omega_v = 5.2/4.5$), the amount of precipitates decreased after reaching a maximum and then leveled off (Figure 5b), which agrees with the overall trend in the entire micromodel.

Dissolution of CaCO_3 Precipitates. CaCO_3 precipitates initially formed near the inlet at high saturation states were found to dissolve over time; Figure 6 compares snapshots of selected pore spaces near the inlet and outlet at three different time points in the high saturation state experiment ($\Omega_c/\Omega_v = 5.2/4.5$). Near the inlet, the width of CaCO_3 precipitates decreased from 50 μm at 11 min to 10 μm at 70 min (Figure 6 left panels). We propose that this reduction occurs because the packing density of precipitates increases over time (i.e., porosity of CaCO_3 precipitates decreases with time), limiting mixing and reaction between solutes. In the pore space near the downgradient location, dissolution was also observed but to a lesser extent: from 23 μm at 29 min to 12 μm at 60 min (Figure 6 right panels).

Permeability Reduction. Changes in permeability as a result of CaCO_3 precipitation were not directly measured in the current study; instead, 3D laser confocal microscopy and fluorescent tracer tests were conducted to qualitatively evaluate permeability reduction. Laser confocal images in selected pores for two experiments ($\Omega_c/\Omega_v = 3.4/2.8$, 4.6/

3.9) show that CaCO_3 precipitates occupied the entire 20 μm depth of the micromodel, which indicates potential blockage of pore space. Crystals formed off-center or those formed along the cylinder walls also occupied the entire 20 μm depth in the micromodel, indicating crystal precipitation was uniform along the vertical dimension in the micromodel. Images of fluorescent tracer in the micromodels from a control experiment (no precipitates in the micromodel) and in the experiments after CaCO_3 precipitation reached pseudo-steady-state were obtained using a 2:1 flow rate of tracer: water. The tracer solution occupied approximately 2/3 of the transverse cross-sectional area of the micromodel in the absence of CaCO_3 precipitates, but only half of the transverse cross-sectional area of the micromodels after CaCO_3 precipitates formed at each of the four saturation states. This confirms blockage of the pore space by CaCO_3 precipitates. Images that support these results are in the Supporting Information (Figures S4 and S5).

Implications. Results from this study indicate mineral precipitation reactions in subsurface sedimentary systems can limit transverse mixing between reactive fluids. Different polymorphs of CaCO_3 crystal precipitation can form along transverse mixing zones depending on the solution saturation state, and these result in a decrease in porosity, a reduction in mixing between reactive species which limits mineral phase transformation, and a reduction in effective permeability. Implications of this study on processes occurring in subsurface sedimentary systems include the following: (i) along the fringes of CO_2 plumes created during geological carbon sequestration mineral precipitation may decrease the effective permeability and result in a buildup of pressure and potential loss of injectivity, (ii) precipitates formed along plume margins during in situ remediation can decrease reactant mixing and decrease cleanup efficiency, (iii) mixing and precipitation of leachate from nuclear waste repository with surrounding groundwater can be an effective way of storing radioactive wastes (26). Experiments performed in this work represent a simplified scenario, more complex solution conditions (i.e., presence of competing cations, different ionic strength, pH, temperature, and pressure) and physical and chemical heterogeneity in 3D porous media are expected. Further study is needed to evaluate how these complexities affect mineral precipitation, and corresponding changes in reactant mixing, mineralogy, porosity, and permeability.

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Supporting Information Available

High resolution image (0.65 μm) used for mineral analysis (Figure S1), total and region specific precipitation versus time from micromodel experiments at intermediate saturation states (Figure S2, S3), laser confocal microscope images of precipitates in selected pores (Figure S4), and images of a fluorescent tracer before and after CaCO_3 precipitation in

the micromodel (Figure S5). This material is available free of charge via the Internet at <http://pubs.acs.org>.

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